CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Third Semester of B. Pharm. Examination
University Theory Examination March - April 2018
PH217/PH229 Pharmaceutical Chemistry - III
Date: 31.03.2018, Saturday   Time: 01:30 p.m. to 04:30 p.m.   Maximum Marks: 80

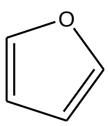
Instructions:

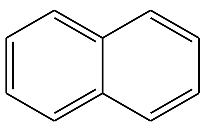
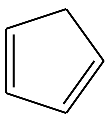
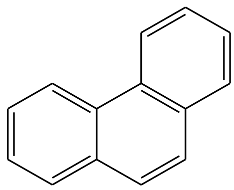
1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

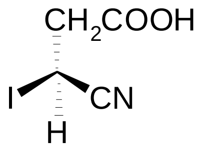
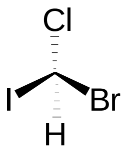
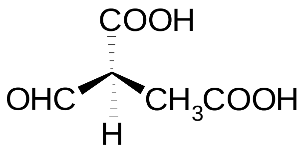
SECTION - I

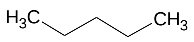
Q 1 Attempt all questions. Each question is of one mark. **20**

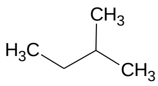
1. Mark the incorrect option about the benzene.
   [A] It gives substitution type of reaction
   [B] In Benzene –C=C- Bond angle is 120^0
   [C] Obey the $(4n+2)\pi$ electrons rule
   [D] All carbons in SP hybridization
2. Electrophilic aromatic substitution reaction in Anthracene most probably occurs at____ position.
   [A] C-1
   [B] C-3
   [C] C-9
   [D] C-2
3. When considering electrophilic aromatic substitution reactions electron withdrawing substituents (e.g. $-SO_3H$) are described as...
   [A] *Ortho/para* directing and activating
   [B] *Ortho/para* directing and deactivating
   [C] *Meta* directing and activating
   [D] *Meta* directing and deactivating
4. All are aromatic structures; Except One

   [A] 

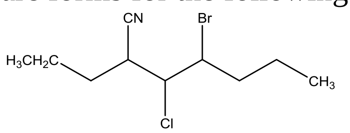
   [B] 
   [C] 
   [D] 
5. Which form is more stable in conformation of n-Butane?
   [A] Skew Staggered
   [B] Skew Eclipsed
   [C] Fully Staggered
   [D] Fully Eclipsed

6. Cis and Trans shows _______________ Configuration.
   [A] Relative
   [B] Absolute
   [C] Optical
   [D] Geometrical
7. Find out the Absolute configuration (R/S) of following structure.
   1.
   
   2.
   
   3.
   
   [A] 1-S,2-R,3-S
   [B] 1-R,2-R,3-R
   [C] 1-R,2-R,3-S
   [D] 1-S,2-S,3-S
8. ___________ Form of Cyclohexane is stable.
   [A] Chair
   [B] Boat
   [C] Half Chair
   [D] Twist Boat
9. Structure A and B is a class of ____________________ isomer


A


B

   [A] Functional
   [B] Chain
   [C] Positional
   [D] Metamerism
10. How many stereoisomers are forms for the following structure?



   [A] 9
   [B] 3
   [C] 8
   [D] 16
11. [3,3'] Sigmatropic rearrangement is an example of _________ type of reaction.
   [A] Cope Rearrangement
   [B] Wolff Rearrangement
   [C] Favorskii rearrangement,
   [D] Lossen rearrangement
12. With Odd no of π Bond, The Cycloaddition Reaction under Photochemical Condition in ____________________ fashion.
   [A] Conroatory
   [B] Disrotatory
   [C] Antrafacial
   [D] Suprafacial
13. All are true about Diel's Alder Reaction.
   [A] The product is Cyclic
   [B] It is a reaction between Diene and Dienophile
   [C] It is a Cycloaddition reaction
   [D] The product has more double bond than diene and dienophile

14. Nucleophilic addition reaction is fast with ___________
    [A] Ether
    [B] Ketone
    [C] Amine
    [D] Alkane
15. Which of the following is the correct statement?
    [A] Aldehyde and Ketone gives substitution type of reaction
    [B] Ether is more reactive towards nucleophilic attack than ester
    [C] Amines is more reactive towards nucleophilic attack than amide
    [D] Nucleophilic substitution reaction is more faster at acyl carbon than saturated carbon
16. Hydrazine Hydrate reagent is used in _______________ type of reaction.
    [A] Wolff Kishner Reduction
    [B] Aldol Condensation
    [C] Reimer Tiemann Reaction
    [D] Clemmensen Reduction
17. Following are the characteristics of the carboxylic acids. Except One
    [A] Dimer formation takes place
    [B] Carboxylic acids have lower boiling point than alcohol
    [C] Electron withdrawing group increase the acidity of it
    [D] Polar in nature
18. _____________ is not derivative of carboxylic acid.
    [A] Acid Chloride
    [B] Ester
    [C] Amine
    [D] Amide
19. All are the examples of Heterogeneous catalyst; EXCEPT One
    [A] Nickel
    [B] Wilkinson's
    [C] Platinum
    [D] Palladium
20. Anchimeric Assistance term associated with ___________ mechanism.
    [A] Nucleophillic Acyl Substitution
    [B] Neighbouring Group Reaction
    [C] Free Radical Substitution Reaction
    [D] Electrophilic Aromatic Substitution Reaction

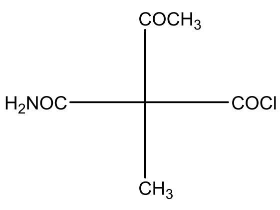
SECTION – II

Q 2 Attempt any TWO of the following

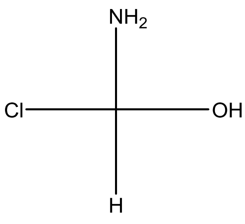
A. Define Following Terms: **05**

i. Huckel's Rule
ii. Erythro and Threo
iii. Racemic mixture
iv. Chiral Compounds
v. D and L

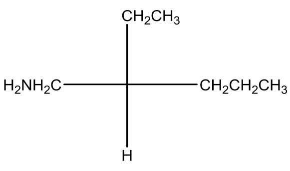
B. (I) Identify the R/S Configuration of following structures. (03 Marks) **05**



(i)

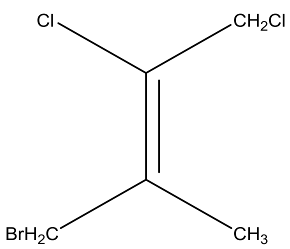


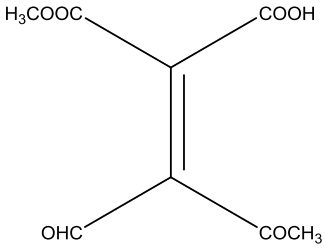
(ii)



(iii)

(II) Identify the E/Z Configuration in following structures. (02 Marks)

(i) 

(ii) 

C. Write a note on Stereochemistry in biphenyl and allene system. **05**

Q 3 **Attempt any <u>TWO</u> of the following**

A. Compare Enantiomers Vs. Diastereomers **05**

B. Give a reason for following statement; (Any two) **05**

i. Amide in more reactive than amine towards nucleophillic attack.
ii. Trans form is more stable than cis form of any geometrical isomer.
iii. Cyclopentadiene is not aromatic but anion of its is aromatic.

C. Write reaction and draw mechanism of Wittig Reaction. **05**

SECTION – III

Q 4 **Attempt any <u>FOUR</u> of the following**

A. Explain nitration of benzene with its mechanism. **05**

B. Write a short note on friedal craft acylation. **05**

C. Explain electrophilic aromatic substitution in naphthalene. **05**

D. Write any three preparations of aldehyde and ketone. **05**

E. Write a short note on kolbe synthesis. **05**

F. Write preparation and reaction of ether. **05**

Q 5 **Attempt any <u>FOUR</u> of the following**

A. Short note on cylcoalkane. **05**

B. Explain 1,2 Vs. 1,4 electrophillic addition of diene. **05**

C. Write preparation and reaction of Phenol. **05**

D. Enlist various techniques to resolve racemic mixture and explain any one on detail. **05**

E. Short note on electrocyclic reaction. **05**

F. Explain neighbouring group mechanism in detail. **05**